Customer Shopping Behaviour Analysis

1. Project Overview

This project analyses customer shopping behaviour using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behaviour to guide strategic

business decisions.

2. Dataset Summary

- Rows: 3,900

- Columns: 18

- Key Features:

- Customer demographics (Age, Gender, Location, Subscription Status)

- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)

- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of

Purchases, Review Rating, Shipping Type)

- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

● **Data Loading**: Imported the dataset using pandas.

● **Initial Exploration**: Used df.info() to check structure and .describe() for

summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900.000000	3900	3900
2	NaN	6	7
No	NaN	PayPal	Every 3 Months
2223	NaN	677	584
NaN	25.351538	NaN	NaN
NaN	14.447125	NaN	NaN
NaN	1.000000	NaN	NaN
NaN	13.000000	NaN	NaN
NaN	25.000000	NaN	NaN
NaN	38.000000	NaN	NaN
NaN	50.000000	NaN	NaN

Missing Data Handling: Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

● **Column Standardization**: Renamed columns to snake case for better readability and documentation.

● **Feature Engineering**:

○ Created age_group column by binning customer ages.

○ Created purchase_frequency_days column from purchase data.

● **Data Consistency Check:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.

● **Database Integration:** Saved the Analysed data in csv format and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

gender	revenue
Male	155438
Female	75191

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67
35	91
37	69
40	60
41	76
43	100
44	69
55	94
57	73
58	64
60	79
62	68
64	79
65	83
67	94

3. **Top 5 Products by Rating** – Found products with the highest average review ratings

	item_purchas...	Average Product Rati...
	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.80
	Skirt	3.79

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_ty...	ROUND(AVG(purchase_amount)...
	Express	60.34
	Standard	58.41

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_sta...	total_custom...	avg_spend	total_revenue
	Yes	1033	59.42	61377
	No	2830	59.81	169252

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchas...	discount_ra...
	Hat	49.67
	Sneakers	49.66
	Coat	48.43
	Sweater	47.20
	Pants	46.43

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

customer_segm...	Number of Custom...
Loyal	3085
Returning	695
New	83

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

item_rank	category	item_purchas...	total_orders
1	Accessories	Jewelry	170
2	Accessories	Belt	159
3	Accessories	Sunglasses	159
1	Clothing	Blouse	169
2	Clothing	Pants	168
3	Clothing	Shirt	168
1	Footwear	Sandals	159
2	Footwear	Shoes	147
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	159

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

subscription_sta...	repeat_buyers
Yes	939
No	2503

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group

age_group	total_revenue
Young Adult	61533
Middle-aged	58380
Adult	55569
Senior	55147

5. **Dashboard in Power BI**

Finally, we built an interactive dashboard in Power BI to present insights visually.

6. **Business Recommendations**

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the "Loyal" segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.